

EXHALE against the published hot-Uranus solution:

Koskinen et al. (2022) Model A at 0.05 au

The acceptance test of the molecular lower-atmosphere line of EXHALE is not our own convergence gate but a published solution of the same problem. Koskinen, Lavvas, Huang, Bourgalais, Ni, Yelle & Sarah (2022, ApJ 929, 52) solved the H/He/H₂ escape of a tidally locked Uranus-like planet at 0.05 au with species transport, and gave the profiles of their Model A. This memo records the inputs that define that model, the EXHALE configuration that reproduces those inputs, the differences of method that remain, and the comparison of the digitized Model A profiles with the EXHALE run benchmarks/koskinen2022_model_a.

1. The target

Model A is defined in section 3.2 of Koskinen et al. (2022): a planet of Uranus mass and radius on a 0.05 au orbit around a Sun-like star, irradiated by the mean solar XUV spectrum of Koskinen et al. (2013a,b) scaled to $F_{\text{XUV}} = 1.6 \text{ W m}^{-2}$ ($1.6 \times 10^3 \text{ erg cm}^{-2} \text{ s}^{-1}$, 13.6 eV–1.24 keV), with H, He and the molecular ions H₂⁺, H₃⁺ and HeH⁺ (their Table 1, reactions P1–P5 and R1–R23), spherical gravity, and a lower boundary at 1 μbar (their Appendix B) where $T = 1140 \text{ K}$ and the mixing ratios are $q_0(\text{H}_2) = 0.84$, $q_0(\text{H}) = 0.026$. Two conventions of theirs are part of the definition: (i) the incident flux is divided by four, a uniform redistribution of the stellar energy over the sphere (their section 3; Koskinen et al. 2013a section 2, “divided the incident stellar flux by a factor of 4”), and (ii) H₃⁺ cools in the optically thin limit. Their published mass-loss rate is $F_M = \rho v r^2 = 1.5 \times 10^9 \text{ g s}^{-1} \text{ sr}^{-1}$, i.e. $\dot{M} = 1.9 \times 10^{10} \text{ g s}^{-1}$ over 4π . The digitized Model A profiles (their Figures 7 and 8, read as described in docs/p23_published_profiles.md sections 4–5) are the reference of every figure and table below; the two rows nearest their lower boundary fall in the near-vertical part of every curve and are order-of-magnitude only.

2. The EXHALE configuration

2.1 Every condition of Model A, and what EXHALE does with it

Checked one by one against the published text (sections 2.5, 3.2, Appendix B, Table 1 of Koskinen et al. 2022; section 2 of Koskinen et al. 2013a for the spectrum; Table 4 of Ribas et al. 2005 for the band fluxes the 1.6 W m^{-2} comes from). “Matched” means the same statement is in force in the EXHALE run; “cannot be matched” says why.

condition	Koskinen et al. (2022) Model A	EXHALE run (benchmarks/koskinen2022_model_a)
gravity	spherical, planet only (Model B adds the Roche potential)	matched : Domain mode: Spherical. The earlier run used the default Roche potential truncated at $L1 = 4.8 R_p$, which moved the sonic point to $2.9 R_p$ and made the velocity 1.3–2.2 times theirs.
planet	Uranus: $8.6813 \times 10^{28} \text{ g}$, lower boundary $3.4249 \times 10^9 \text{ cm}$	matched : $0.045739 M_J$, $0.4899 R_J$ (0.9992 and 1.0002 of theirs).
orbit, star	0.05 au, Sun-like	matched : 0.05 au, $1 M_\odot$.

condition	Model A	EXHALE run (continued)
domain	1.34–9.7 R_p = 1–7.24 lower-boundary radii, 410 stretched levels	matched: Outer radius [R_p]: 7.24; 500 cells (Mixed grid, 50 uniform base cells).
lower boundary	$p = 1 \mu\text{bar}$, $T = 1140 \text{ K}$, $q_0(\text{H}_2) = 0.84$, $q_0(\text{H}) = 0.026$; v_0 from the flux constant	matched: base.inp (p_base 1e-6, T_base 1140, q_H2_base 0.84); the base velocity follows the mass flux.
upper boundary	extrapolated outflow (Tian et al. 2005)	matched: outflow.
XUV flux	1.6 W m^{-2} over 0.1–91.1 nm at 0.05 au (Ribas et al. 2005)	matched: the spectrum file integrates to $1600 \text{ erg cm}^{-2} \text{ s}^{-1}$ over 5–911 Å.
spectrum	“mean solar” (SOLAR2000 + Woods & Rottman; Koskinen et al. 2013a)	matched in the band totals, not in detail: no copy of their spectrum exists here. The Gueymard (2003) composite (VULCAN’s Gueymard_solar.txt, a solar-surface flux) was rescaled band by band to the Ribas et al. (2005) Sun values (1–20, 20–100, 100–360, 360–920, 920–1180 Å: 0.15, 0.70, 2.05, 1.00, 0.74 $\text{erg cm}^{-2} \text{ s}^{-1}$ at 1 au); the composite’s 1–100 Å was 2.4–3.1 times too weak and its 100–360 Å 1.5 times too strong. The shape inside a band changes the optically thin heating per particle by 7 per cent (checked against the p-winds solar file). The file has nothing below 5 Å.
dayside average	incident flux divided by 4	matched: Rate/4 + Mdot.
photoelectrons	100 per cent of $h\nu - I$ heats; I is lost in recombination or chemistry	matched: Secondary_ionization: False, He_rec_coupling: False, Molecular reaction heat: False. (Section 4: their Figure 9 heating rate is 2.5–3 times what this accounting gives; Photoelectron heating: full is the comparison option that tests the alternative.)
species	H_2 , H, He, H_2^+ , H_3^+ , H^+ , He^+ , HeH^+	matched: Include He23S? False; EXHALE also carries He^{++} (negligible here).
reactions	Table 1: P1–P5, R1–R23	matched: R5–R23 have always been the Table 1 fits (mol_rates.f90); R1–R4 (H^+ , He^+ recombination; H, He collisional ionization) by Atomic rate set: Koskinen2022 (R3, R4 were already EXHALE’s default). Cross sections: Verner et al. (1996) here against Hummer (1963), Yan et al. (1998).
H_2 photodissociation	not included (postponed)	matched: Stellar LW flux: 0.0 (a stated zero now wins over the value a loaded spectrum would supply).
H_2 photoionization channels	P3–P5 (Yan 1998; Chung 1993)	matched: the same channel set (section 151).
cooling	recombination, H_3^+ (optically thin, non-LTE factor of Koskinen et al. 2009), H $\text{Ly}\alpha$ (Glover & Jappsen 2007)	partly: recombination and $\text{Ly}\alpha$ the same; H_3^+ with the Miller et al. (2013) Table 6 non-LTE factor – their Figure 9 implies $\sim 3 \times 10^{-13} \text{ erg s}^{-1}$ per molecule at 3 r_{base} against our 1.1×10^{-12} .
conduction, viscosity	both in the equations; conduction “largely negligible”	conduction matched (Conduction: True, Watson et al. 1981); viscosity off here (their coefficient is not stated; negligible in the wind).

condition	Model A	EXHALE run (continued)
diffusion	every species advected and diffused; K_{zz} value not stated	cannot be matched exactly: H_2 and H^+ are carried by the flow (molecular diffusion of H_2 , $K_{zz_base} = 1e9$); He, He^+ , H_2^+ , H_3^+ , HeH^+ are the local equilibrium of each cell.
energy equation	$u = c_v T$, c_v not stated	tested both: the H_2 rovibrational ladder (default) and Caloric EOS: monatomic; the outer temperature does not depend on it (section 4).
numerics	operator-split van Leer, 1 s steps, Shapiro filter	not matched (different method); the steady state should not depend on it.

2.2 The runs

benchmarks/koskinen2022_model_a/ holds the input files and the final states. matched_hnu_minus_I/ is the run with every condition of the table above in force and the physical photoelectron accounting ($h\nu - I$); matched_full_hnu/ is the same run continued with Photoelectron heating: full, and matched_fraction_0p55/ with a stated fraction 0.55 of the photon energy per ionization (section 4; both are comparison options, not physics). Both march (no Solver key: the steady solver would put the composition back on its local root, section 168 of the changelog) for the step count set by EXHALE_MAXSTEPS; du_th [PLM,WENO3]: $1.0e9 \ 1.0e-3$ starts the march on WENO3 at once, and the flux-spread stop, which the breathing base never lets trip, would only end the march. The matched run was started by interpolating the earlier Roche-domain state onto the spherical 7.24-radius grid (interp_ic.py of the session scratch, element ratios and the mass closure exact by construction) and marched 2×10^5 steps with the Gueymard-shaped spectrum, then 4×10^5 with the Ribas-band spectrum; the $h\nu$ run is 6×10^4 steps from the 2×10^5 point of the latter, and the 0.55 $h\nu$ run (matched_fraction_0p55/, Photoelectron heating: 0.55) 1.2×10^5 steps from the 4×10^5 point; a 0.65 run, 1.2×10^5 steps, is not kept (its outer temperature was 175–465 K above Model A's, and it was the same at 6×10^4 and 1.2×10^5 steps, which is how the outer wind's settling time was measured). The base of the matched run is still settling: the base mass flux is 1.9 times the wind's (2.2 at 2×10^5), the layer's H_2 density at $1.10 r_{base}$ rose from 3.3 to 4.8×10^{10} over the first 2×10^5 steps (Model A: 7.2×10^{10}) and the 1.05 temperature from 841 to 1090 K by 4×10^5 (1096 at 1.045 in Model A). The layer's own time scale is far longer than any march.

2.3 Differences of method that remain

After the table of section 2.1 these are the differences the runs still carry, none of which the paper's text lets us remove:

- **Species transport.** Model A advects and diffuses every species. EXHALE carries H_2 and H^+ with the flow and leaves He, He^+ , H_2^+ , H_3^+ and HeH^+ at the local equilibrium of each cell. For H/H^+ this is the regime that matters ($Pr/|v| = 0.15\text{--}0.35$ above $1.5 r_{base}$, section 168.7 of the changelog) and it is carried; for He^+ the local root overstates the ion by a factor of about three against their curve.
- **Eddy diffusion.** Their K_{zz} is not stated; ours is $10^9 \text{ cm}^2 \text{ s}^{-1}$ at the base.
- **H_3^+ non-LTE factor.** Theirs from the detailed-balance table of Koskinen et al. (2009); ours the Miller et al. (2013)

Table 6 factor. Their Figure 9 implies about 3×10^{-13} erg s⁻¹ per molecule at $3 r_{\text{base}}$ against our 1.1×10^{-12} : their volumetric cooling there is larger than ours only because their H_3^+ density is.

- **Heat capacity.** Their c_v is not stated; ours is tested both ways (section 4) and does not decide anything here.
- **Viscosity** (their coefficient not stated; ours off) and the **numerical method** (operator splitting with a Shapiro filter against our Godunov scheme); neither should move a steady state.
- **The heating per ionization** – the one difference that decides the outer wind, section 4.

3. Comparison

Table 2 lists the four profiles of the Koskinen comparison at eight radii for the three photoelectron accountings, and Figures 1–4 show the full curves.

Table 2: EXHALE against Model A. $f(\text{H}_2) = 2n(\text{H}_2)/n_{\text{H}}$; n_e is the sum of the ionic charges; the $h\nu - I$, $h\nu$ and $0.55 h\nu$ columns are the three photoelectron accountings of section 4. Model A values are interpolated from the digitized curves.

r/r_{base}	$f(\text{H}_2)$				$n(\text{H}_3^+) [\text{cm}^{-3}]$				$T [\text{K}]$				$n_e [\text{cm}^{-3}]$			
	$h\nu - I$	$h\nu$	$0.55 h\nu$	A	$h\nu - I$	$h\nu$	$0.55 h\nu$	A	$h\nu - I$	$h\nu$	$0.55 h\nu$	A	$h\nu - I$	$h\nu$	$0.55 h\nu$	A
1.05	0.976	0.975	0.976	0.982	8.1×10^2	8.0×10^2	8.2×10^2	3.1×10^3	1090	1019	1103	1180	1.3×10^7	1.3×10^7	1.3×10^7	5.0×10^6
1.10	0.964	0.961	0.966	0.976	2.3×10^3	2.2×10^3	1.6×10^3	5.0×10^3	1507	1512	1452	1570	1.0×10^7	1.3×10^7	1.2×10^7	4.4×10^6
1.15	0.934	0.931	0.938	0.963	5.2×10^3	7.2×10^3	4.0×10^3	7.0×10^3	1753	1829	1707	1720	1.2×10^7	1.0×10^7	1.6×10^7	7.2×10^6
1.20	0.882	0.887	0.891	0.936	6.0×10^3	8.6×10^3	5.3×10^3	9.1×10^3	1925	2016	1898	1900	1.5×10^7	1.2×10^7	1.8×10^7	9.6×10^6
1.30	0.761	0.790	0.781	0.843	4.4×10^3	7.3×10^3	4.5×10^3	1.2×10^4	2175	2307	2177	2270	1.9×10^7	1.4×10^7	2.0×10^7	1.0×10^7
1.50	0.587	0.644	0.620	0.688	2.5×10^3	6.8×10^3	3.7×10^3	1.1×10^4	2656	2985	2790	2870	1.4×10^6	8.2×10^6	1.1×10^7	5.7×10^6
2.00	0.408	0.445	0.433	0.508	9.9×10^2	7.3×10^3	2.7×10^3	7.2×10^3	3461	4830	4119	4120	4.5×10^6	1.9×10^6	2.9×10^6	1.8×10^6
3.00	0.311	0.296	0.322	0.403	1.8×10^2	1.8×10^3	6.7×10^2	1.9×10^3	3479	6389	4840	4740	1.8×10^6	8.2×10^5	1.1×10^6	7.8×10^5

4. What the matching found, in order

Gravity. The single largest difference was the potential. The earlier run used EXHALE’s default Roche domain (the stellar tidal term, truncated at $L1 = 4.8 R_p$), which puts the sonic point at $2.9 R_p$ against Model A’s $4.7 R_p$ (isothermal) and makes the velocity 1.3–2.2 times theirs and the density 0.4–0.7 times. With Domain mode: Spherical and the domain extended to 7.24 base radii the velocity fell onto their curve and the neutral densities into the size of the plotted points.

Spectrum. The composite solar spectrum first used had 2.4–3.1 times too little 1–100 Å flux and 1.5 times too much 100–360 Å flux against the Ribas et al. (2005) Sun values the paper’s 1.6 W m^{-2} comes from; rescaled band by band, the deep layer warmed (the $1.05 r_{\text{base}}$ trough from 841 to 976 K) and the outer wind cooled (3650 to 3470 K at $2 r_{\text{base}}$). The shape inside a band moves the optically thin heating per particle by 7 per cent.

What is left is the heating rate itself. With every condition of section 2 in force, the run reproduces the peak of the stellar heating ($1.9 \times 10^{-8} \text{ W m}^{-3}$ at 1.16 against their 1.8×10^{-8} at $1.24 r_{\text{base}}$) but not its width: their Figure 9 (digitized, model_a_fig9_digitized.txt) shows 2.5–3 times our heating from $1.5 r_{\text{base}}$ outward and 2.3 times ours in the deep

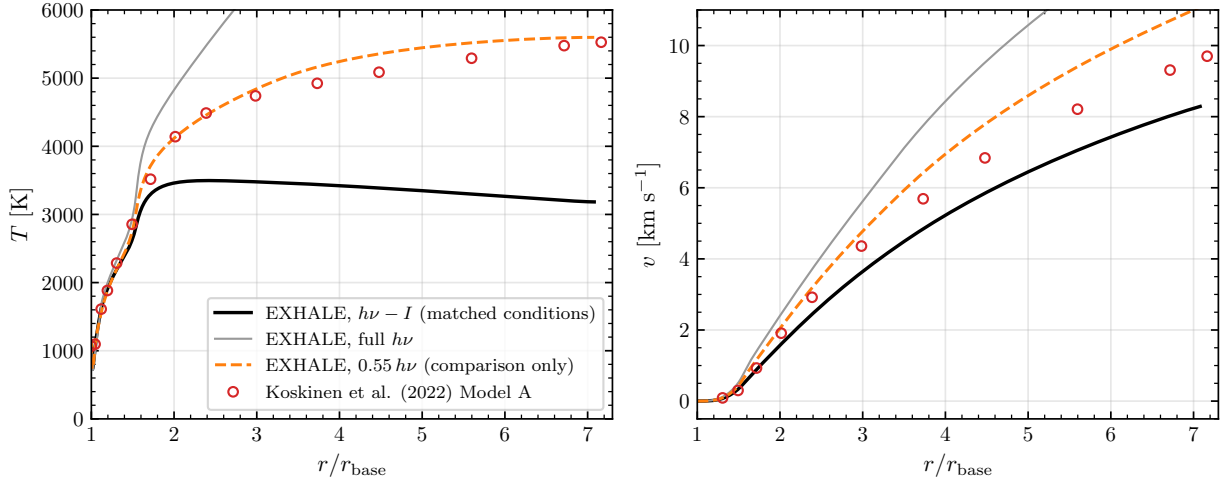


Figure 1: Temperature and velocity. Circles: Model A digitized from their Figure 7. Black: the matched run with $h\nu - I$; gray: the whole photon energy deposited (Photoelectron heating: full); dashed: a stated 0.55 of it (comparison only).

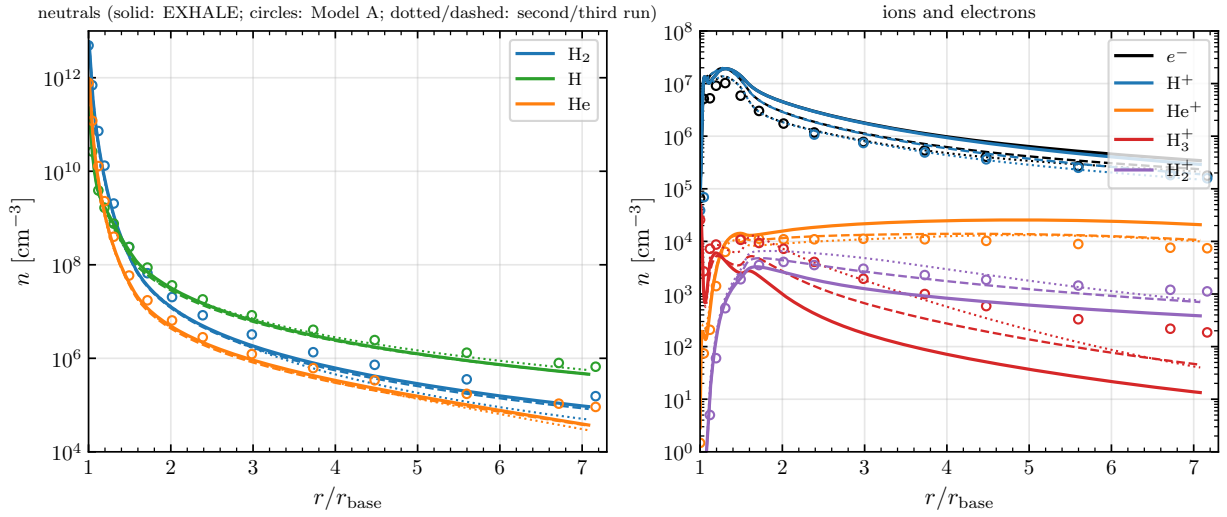


Figure 2: Number densities of the neutrals (left) and of the ions and electrons (right); circles are the Model A readings of their Figure 8 (H⁺ circles absent where hidden under the electron curve). Solid: $h\nu - I$; dotted: full $h\nu$; dashed: $0.55 h\nu$.

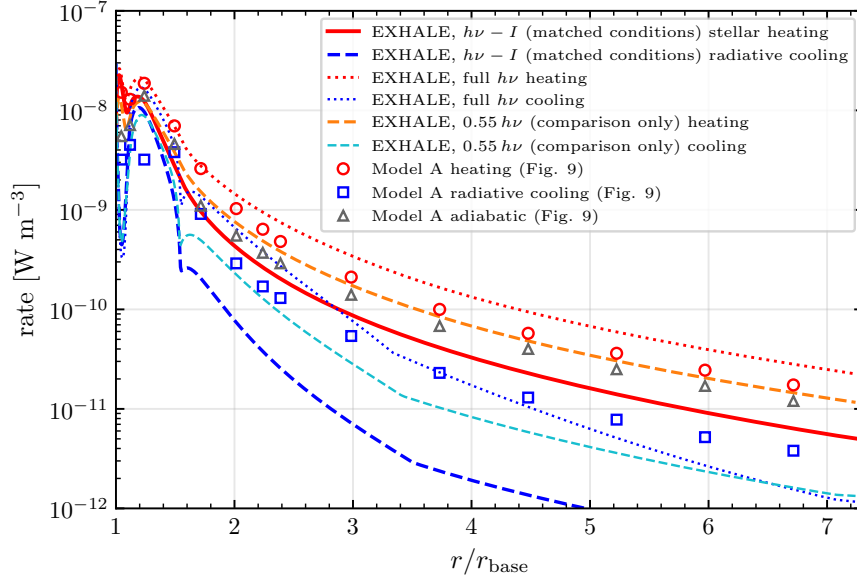


Figure 3: Stellar heating and radiative cooling rates against the digitized Figure 9 of Koskinen et al. (2022): circles their heating, squares their radiative cooling, triangles their adiabatic term. Solid/dashed red and blue: the matched run with $h\nu - I$; dotted: with the full photon energy; dashed orange/cyan: with 0.55 of it.

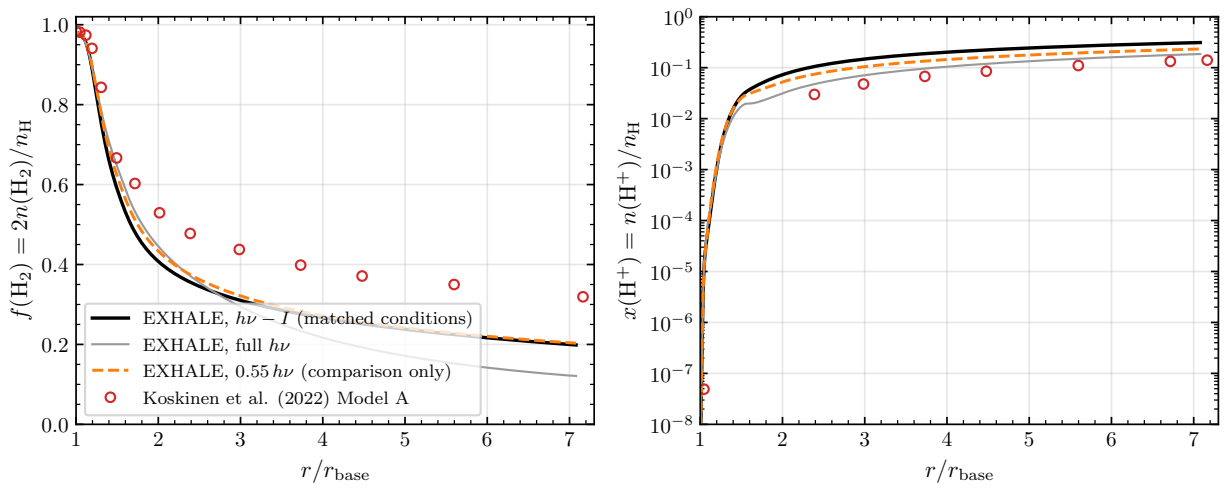


Figure 4: Fraction of H nuclei bound in H_2 (left) and ionized fraction of hydrogen (right); black $h\nu - I$, gray full $h\nu$, dashed 0.55 $h\nu$.

layer, at the same ionization rate (their Figure 15 gives $8 \times 10^{-6} \text{ s}^{-1}$ per H atom at $2 r_{\text{base}}$ against our 1.0×10^{-5}) and a column above the point that is larger, not smaller, than ours. So each of their ionizations deposits 2.5–3 times more heat than $h\nu - I$ gives for a spectrum with the Ribas band fluxes (3.2 eV per H ionization, mostly from the Lyman continuum), and no spectrum with those band totals changes that by more than 7 per cent. The steady energy balance makes the consequence explicit: with their own digitized T and v the adiabatic term alone is $8\text{--}10 \times 10^{-17} \text{ erg s}^{-1}$ per particle at $2\text{--}3 r_{\text{base}}$, and their temperature still rises there, while our heating per particle is 7×10^{-17} and falls. The heat capacity does not enter that sign (Caloric EOS: monatomic changed the outer temperature by 20 K), and neither does the H_3^+ cooling, which is 10–25 per cent of the heating in both models there.

The bracket. Photoelectron heating: full deposits the whole photon energy per ionization. That run overshoots their heating by 1.5 (Figure 3) and their temperature by 700–1650 K, but its electron density, H_3^+ density and radiative cooling fall on Model A’s curves (Table 2, Figures 2 and 3): $n_e = 1.9 \times 10^6$ against 1.8×10^6 at 2.0, 8.2×10^5 against 7.8×10^5 at 3.0; H_3^+ 7.3×10^3 against 7.2×10^3 and 1.8×10^3 against 1.9×10^3 . The physical accounting undershoots the heating by 0.4 and the temperature by 650–1270 K, with twice their electron density. Model A lies between the two, at about 0.65 of the photon energy per ionization. Nothing in the paper’s text yields that number: it states $h\nu - I$ with I lost to recombination or chemistry, and a mean solar spectrum normalized to Ribas et al. (2005).

The fraction run. With a stated 0.55 of the photon energy per ionization (Photoelectron heating: 0.55, a comparison option that claims no physics) the stellar heating rate falls on their Figure 9 from $1.5 r_{\text{base}}$ outward (Figure 3), and the temperature on their Figure 7 within the size of the points: 4119 K at 2.0 (theirs 4120), 4840 at 3.0 (4740), 5600 at 7 (5527); the velocity is 10 per cent above theirs at $3\text{--}6 r_{\text{base}}$. The composition does not follow all the way: n_e stays 1.4–1.6 times theirs at $1.5\text{--}3.0$, H_3^+ 0.35–0.5 of theirs, He^+ 3 times theirs, and $f(\text{H}_2)$ 0.06–0.08 below theirs above $1.5 r_{\text{base}}$ (Table 2). Section 4.1 says why.

Layer. Below $1.3 r_{\text{base}}$ all three runs are within about 100 K of Model A, the 1.05 trough included by 4×10^5 steps (1090–1109 K against 1180); the electron density there is 1.5–3 times theirs, feeding the R10 proton sink that sets their low H^+ .

What would settle it. The authors’ spectrum file and the heating rate per ionization their code deposits (or the tabulated Figure 9 curves).

4.1 Why the results are not identical: every difference, with its size

In the order of what it does to the comparison.

1. **The heat deposited per ionization.** Model A’s stellar heating (Figure 9, digitized) is 2.5–3 times the $h\nu - I$ rate at the same ionization rate (Figure 15) and at a larger neutral column above the point. Any spectrum with the Ribas band fluxes gives 3.2 eV per H ionization; their profile needs about 8 eV, i.e. 0.55–0.65 of the photon energy, and no statement in the paper produces that number. Effect: with $h\nu - I$ the outer temperature is 650–1270 K low, the velocity 20 per cent low, the electron density 2.3–2.5 times high and H_3^+ 8–12 times low; with the fraction 0.55 the temperature and heating match. This is the difference that decides the outer wind. It is a property of their code, not of the physics stated in the paper, and it cannot be removed here without the tuning above.
2. **Species transport.** Model A advects and diffuses every species (their Equation B1 with the diffusion velocities

of B9); EXHALE carries H_2 and H^+ and puts He , He^+ , H_2^+ , H_3^+ and HeH^+ at the local equilibrium of each cell. For He^+ this is what a local root does where $Pr/|v| < 1$ – it overstates the ion, here by a factor of about three above $1.5 r_{base}$ (the same 5–10 times excess H^+ had before it was transported, section 168). For H_3^+ the local root is adequate (its chemistry is fast) but its supply, H_2^+ from H_2 , and its sink, dissociative recombination with n_e , both carry the errors of items 1 and 3. Removing this difference means carrying He^+ (and the molecular ions) as transported carriers; the operator takes any species, and the proton is the template.

3. **The layer is not at its steady state.** The base mass flux is still 1.9 times the wind's after 4×10^5 steps and falling slowly, the H_2 density of the layer is 0.7 of theirs at 1.10–1.30, and the 1.05 trough sits 70–90 K below. The layer's own time scale is of order 10^7 – 10^8 s, which no march reaches at a base time step of 2 s, and the steady solver that could close it cannot yet hold a transported H^+ (the proton row is not a Newton unknown; section 171). Effect: a weaker R10 sink, hence 1.5–3 times their electron density in the layer and, carried outward, part of the outer excess; the mass-loss rate $10^{10.10}$ – $10^{10.16}$ g s⁻¹ against their $10^{10.28}$ follows from the same under-filled layer.
4. **The H_2 fraction of the wind** is 0.06–0.10 below theirs above $1.5 r_{base}$ in all three runs. Its per-molecule loss rate at 1.5 agrees with theirs (1.0×10^{-5} against 1.2×10^{-5} s⁻¹, photoionization plus R10), so the deficit is in the supply from the layer (item 3) and in the electron density that feeds R10 (items 1–3), not in the H_2 chemistry.
5. **H_3^+ cooling per molecule.** Theirs, from the Koskinen et al. (2009) detailed-balance table, is about 3×10^{-13} erg s⁻¹ at $3 r_{base}$ (their Figure 9 cooling over their Figure 8 density); ours, with the Miller et al. (2013) Table 6 factor, 1.1×10^{-12} . Their volumetric cooling is nevertheless larger because their H_3^+ density is. Effect on the temperature: 10–25 per cent of the heating in both models, second order next to item 1.
6. **The spectrum's shape inside the Ribas bands** is not theirs (SOLAR2000 with the strong lines resolved against a smoothed composite): 7 per cent in the optically thin heating per particle, and the file has no flux below 5 Å (the 1–5 Å part of the 1–20 Å band is carried by 5–20 Å), which touches only the deepest cells.
7. **Cross sections.** Verner et al. (1996) for H and He against Hummer (1963) and Yan et al. (1998): a few per cent near threshold.
8. **Eddy diffusion** (K_{zz} not stated; 10^9 cm² s⁻¹ here), **viscosity** (coefficient not stated; off here) and the **heat capacity** (c_v not stated): the last was tested and moves the outer temperature by 20 K; the first two act only in the layer.
9. **Numerics.** 500 cells against 410, a Godunov scheme against operator splitting with a Shapiro filter, and our marching end point against their steady state; a steady state should not depend on these, but item 3 says ours is not one in the layer.
10. **The reading of their figures.** About 0.1 dex in the densities and rates, 100 K in the temperature, and the two rows nearest their lower boundary are order-of-magnitude only (p23_published_profiles.md section 4); the deep-layer electron density they show violates their own charge balance and is not used.

Items 1 and 3 are the ones that keep the results from being identical; item 2 is the next; the rest are at the level of the points.

5. Provenance

Figures and the table: docs/k22_model_a_figures.py (run on benchmarks/koskinen2022_model_a/output). Digitized Model A: docs/p23_published_profiles.md section 5.1. The electron-density diagnosis that led to the configuration: docs/k22_electron_density_excess.md; the changes of the code behind it: docs/Update_EXHALE.md sections 168–170.